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# Discussion Assignment 5

**Sugarloaf Peak** MATH 2210, Spring 2018

## Background

Thinking way back to Calculus 1 you may recall that derivatives can be used to find maximum and minimum values of functions of a single variable. When those techniques are extended to two variables, we discover both similarities and differences. When working with single variable functions we use the term critical points to describe candidates for local extrema. Recall that a critical point, $c$ of a function $g(x)$, is defined to be a point in the domain of $g(x)$ at which $g'(c) = 0$ or $g'(c)$ does not exist. Extending this definition to two dimensions we say that a point $(a, b)$ in the domain of $g(x, y)$ is a critical point of $g(x, y)$ if $\nabla g(a, b) = \mathbf{0}$ or at least one of the partial derivatives $g_x$ or $g_y$ does not exist at $(a, b)$. Noting that since $\nabla g(x, y)$ is a vector we find that the determination of critical points can be a difficult task if $g(x, y)$ is a nonlinear function. In this project we will explore one technique for solving nonlinear systems of equation in two variables. Specifically, you will determine how the linear approximation equation can be used to derive Newtons method for solving nonlinear equations.

In lecture we have seen that if a function $f$ is differentiable at $(a, b)$ then the linear approximation to the surface $z = f(x, y)$ at the point $(a, b, f(a, b))$ is given by

$$L(x, y) = f_x(a, b)(x - a) + f_y(a, b)(y - b) + f(a, b).$$

An important application of linear approximations of functions is Newtons methods for solving equations of the form

$$\begin{aligned} f_1(x, y) &= 0 \\ f_2(x, y) &= 0. \end{aligned}$$

The idea is to linearize both equation system and solve for $(x, y)$, repeating the steps as many times as is necessary.

Start with an initial guess that is sufficiently close to the roots of $f_1(x, y)$ and $f_2(x, y)$. By replacing $f_1(x, y)$ and $f_2(x, y)$ with their linear approximations at the point $(x_0, y_0)$ and solving the (simpler) equations $L_1(x, y) = 0$ and $L_2(x, y) = 0$ simultaneously for $x$ and $y$ a better estimate of the root can be obtained. **As part of your report it is required that you write out these linear approximations and show that we want to find $x$ and $y$ such that**

$$\begin{aligned} \frac{\partial f_1}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f_1}{\partial y}(x_0, y_0)(y - y_0) &= -f_1(x_0, y_0) \\ \frac{\partial f_2}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f_2}{\partial y}(x_0, y_0)(y - y_0) &= -f_2(x_0, y_0). \end{aligned}$$

Recall that there are several ways (substitution and elimination are popular) to solve two linear equations with two unknowns. The simplest way symbolically is to use matrices. If we define the **Jacobian matrix**

$$J(\boldsymbol{x}_0) = \begin{bmatrix} \frac{\partial f_1}{\partial x}(\boldsymbol{x}_0) & \frac{\partial f_1}{\partial y}(\boldsymbol{x}_0) \\ \frac{\partial f_2}{\partial x}(\boldsymbol{x}_0) & \frac{\partial f_2}{\partial y}(\boldsymbol{x}_0) \end{bmatrix}$$

where $\boldsymbol{x}_0$ represents the point $(x_0, y_0)$, the above equations can be written as

$$J(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0) = -\boldsymbol{f}(\boldsymbol{x}_0)$$

which has the solution

$$\boldsymbol{x} = \boldsymbol{x}_0 - J^{-1}(\boldsymbol{x}_0)\boldsymbol{f}(\boldsymbol{x}_0).$$

Here the matrix $J^{-1}(\boldsymbol{x}_0)$ is called the **inverse** of the matrix $J(\boldsymbol{x}_0)$ and $\boldsymbol{f}(\boldsymbol{x}_0) = \begin{bmatrix} f_1(\boldsymbol{x}_0) \\ f_2(\boldsymbol{y}_0) \end{bmatrix}$. In general, Newtons method is defined by the iteration

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n - J^{-1}(\boldsymbol{x}_n)\boldsymbol{f}(\boldsymbol{x}_n).$$

Note that you do not need to use the Jacobi matrix to solve this problem. Substitution and elimination are entirely acceptable techniques for solving the problem at hand. The previous discussion is only meant to introduce you to the concept of the Jacobian since it plays a prominent role in the second half of this course.

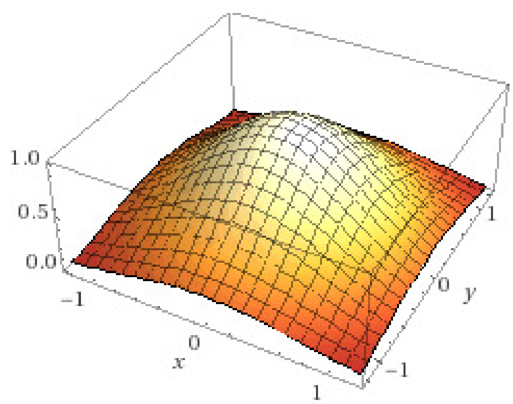
## Question

Using sophisticated GPS technologies and advanced approximation theory it has been found that the surface of Sugarloaf Mountain can be represented with the function

$$s(x, y) = e^{-x^2-y^2}$$

Here the point $(x, y, s(x, y)) = (0, 0, s(0, 0))$ represents the location of Sugarloaf campground. Then for a position $(x, y)$ measured in miles away from the Sugarloaf campground the function $s(x, y)$ gives the elevation (in miles) relative to Sugarloaf campground. Your task is to accurately determine the position of the peak of Sugarloaf mountain. In order to accomplish this task you are required

(a)




(b)

Figure 1: Sugarloaf Mountain and its surface representation as the function $s(x, y)$.

to use Newtons method to find the critical points of $s(x, y)$. From Figure 1b it is reasonable to assume that the peak is near the position $(1, 2)$. Use the point $(1, 2)$ as an initial guess for Newtons method and stop the iteration when your estimate is accurate up to two decimals points.

(a) The peak is located at the point $(x, y) = (0.533, 2.54)$.

(b) The peak is located at the point $(x, y) = (1.18, 1.83)$.

(c) The peak is located at the point $(x, y) = (1.16, 1.84)$.

(d) The peak is located at the point $(x, y) = (1.14, 1.85)$.

## Group Work

The group work portion of your grade is worth 3 points. As a group you are allowed to submit three answers. However, incorrect answers will be counted as wrong. For example, if the correct answer is $(d)$ and your group submits $(a), (d), (d)$ as their answers the score will be 2/3 on this portion of the assignment.

## Individual Work

Now you are to write a report, one page maximum, that convinces an audience of your peers that you understand how to use Newtons method to estimate the location of Sugarloaf peak. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing good mathematics is posted on our WyoCourses page.